AU 2024
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SD2336

# Spit It Out: How AEC and the Manufacturing Data Model API Can Feed ERP Systems

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## Learning Objectives

- Learn how to extract product and quantity information from Revit data using the AEC Data Model API
- Learn how to extract bill of material information from Fusion CAD data using the Manufacturing Data Model API
- Explore the limitations and learn about possible workarounds to fully automate synchronizations

## Description

Autodesk Construction Cloud (ACC) or Fusion Team is used to store product data created with Revit, Fusion, Inventor or other Autodesk CAD authoring tools. The data stored in these cloud systems is often the most important intellectual property and is also critical for users who do not have access to Autodesk cloud-based systems. Learn how you can use Autodesk's AEC Data Model API and Manufacturing Data Model API to extract BOMs, product quantities and other critical metadata and publish it to ERP- or other enterprise-systems. Accurate, reliable and without manual data entry!

## Speaker(s)

Christian Gessner is a co-founder and Head of Research & Innovation at COOLORANGE. In this role, he drives research into cutting-edge technologies that enable customers to effectively automate, implement, and customize Autodesk CAD, PDM, and PLM solutions, ensuring seamless integration with enterprise systems. With over 25 years of experience in full-stack software development, Christian specializes in Autodesk product data and lifecycle management and Microsoft development technologies. Before founding COOLORANGE, he was a member of the data management software engineering team at Autodesk.

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## Purpose of the document

The AU class "*Spit It Out: How AEC and the Manufacturing Data Model API Can Feed ERP Systems*" focuses on Autodesk's Manufacturing and AEC Data Model APIs. It compares the two GraphQL APIs to other vendors' APIs to highlight the differences.

Two customer projects are used to demonstrate the APIs in action. The first project uses the Autodesk Manufacturing Data Model API to transfer Item- and BOM-information from Autodesk Fusion to the ERP system "Microsoft Business Central".

The second project uses the Autodesk AEC Data Model API to extract component information from Revit files stored in the Autodesk Construction Cloud. The extracted information is used to create "products" and "deliveries" in the ERP system "Odoo".

This document describes the configuration of the sample code provided with this class as well as the API-related configuration of the ERP systems "Microsoft Business Central" and "Odoo".

The step-by-step instructions in this document enable developers to re-use and debug the sample code that can be found in the following GitHub repository:
[https://github.com/christiangessner/AU2024_SD2336](https://github.com/christiangessner/AU2024_SD2336)

## Understanding GraphQL: The Technology Behind Autodesk APS

GraphQL

### What is GraphQL?

- Developed by Meta (formerly Facebook) in 2012 and open-sourced in 2015, GraphQL is a flexible query language for APIs that allows clients to request specific data without the need for multiple endpoints.
- Key Features
  - Precise data fetching: Clients request exactly the data they need.
  - Single endpoint: Unlike REST APIs, GraphQL uses a single endpoint to manage various queries.

### How Autodesk APS Leverages GraphQL

- Autodesk APS Data Model APIs use GraphQL to manage AEC and Manufacturing Data Models, allowing efficient, granular access to data from BIM and CAD models.

### Benefits of GraphQL

- Reduced API Complexity: With a single endpoint, GraphQL allows for querying deeply nested models in AEC and Manufacturing data without multiple round trips.
- Customizable Queries: Clients can tailor data requests to match their exact needs, reducing over-fetching typical in REST or OData.

### GraphQL - Wikipedia

***GraphQL** is a data query and manipulation language for APIs, that allows a client to specify what data it needs ("declarative data fetching"). A GraphQL server can fetch data from separate sources for a single client query and present the results in a unified graph,[2] so it is not tied to any specific database or storage engine.*
*The associated GraphQL runtime engine is open-source.*

### GraphQL - Useful Links

https://graphql.org/
https://github.com/graphql
https://learning.postman.com/docs/sending-requests/graphql/graphql-overview/
https://learning.postman.com/docs/sending-requests/graphql/graphql-http/

### Autodesk Data Model APIs - Useful Links

**AU class: Access Granular Design Data Using GraphQL in Autodesk Platform Services**
https://www.autodesk.com/autodesk-university/class/Access-Granular-Design-Data-Using-GraphQL-in-Autodesk-Platform-Services-2023

**Author:** Augusto Goncalves
**Description:** In this class, we will go over all the latest details about the Autodesk Platform Services Data Initiatives from an API perspective. Access your design data in a granular fashion using GraphQL in Autodesk Platform Services. If you're a customer, developer, or partner and want to see the latest information about the API status for each of the three platform data areas, then this is the class for you. As you have heard before, Autodesk is investing heavily in cloud data with Autodesk Fusion 360 (manufacturing), Autodesk Forma (AEC /BIM), and Autodesk Flow (media and entertainment) industry groups. At the center is also the Data Exchange service that allows data to be shared in a specific and durable way. From authoring to access, this class will explain the details about each service and how to use the API to access the data. We'll also cover the fact that data can be accessed across each of the information models, and via Data Exchange, in an easy fashion.

#### Manufacturing and AEC Data Model Explorer

After authenticating with APS, the Manufacturing and AEC Data Model Explorer enable developers to easily execute GraphQL queries. Both tools integrate a GraphiQL Data Explorer and the integrated GraphQL Voyager tool helps to analyze and understand the APIs data structures:
https://mfgdatamodel-explorer.autodesk.io/
https://aecdatamodel-explorer.autodesk.io/

#### Official API Documentation

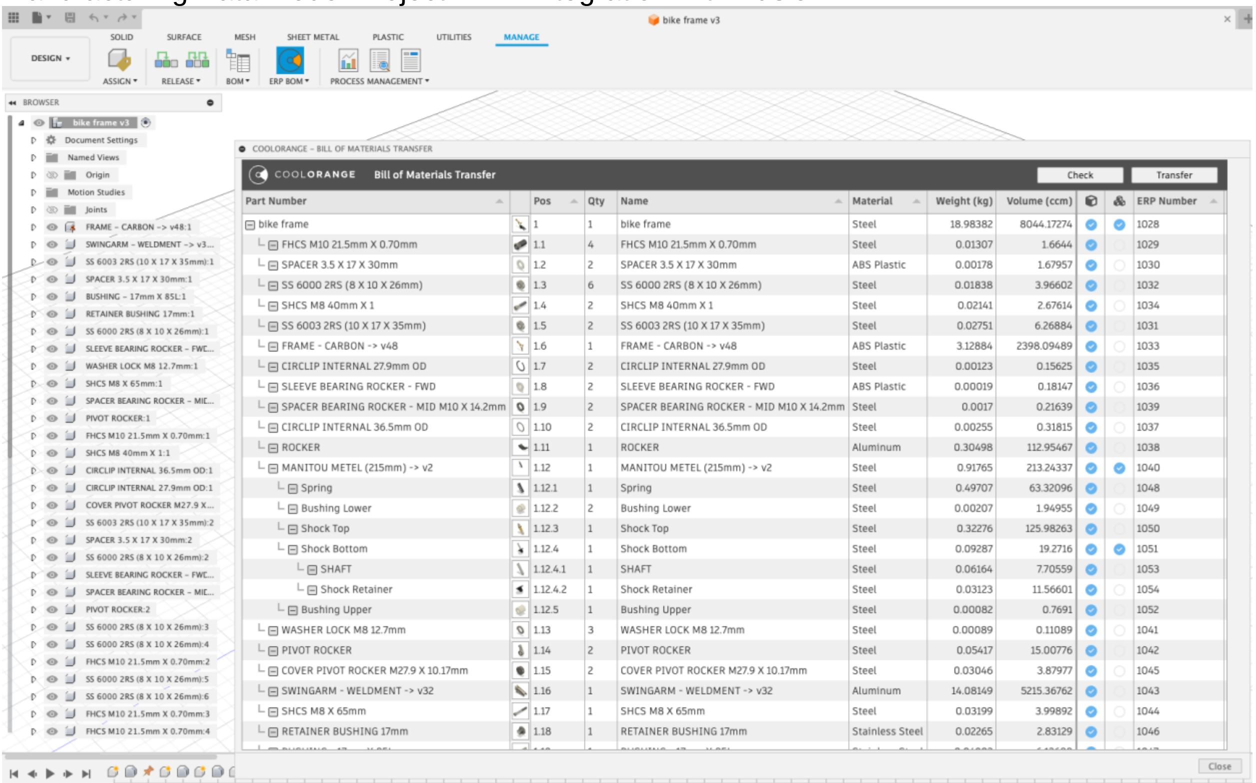
Beside an API reference for every all Autodesk Cloud APIs, the Autodesk APS documentation provides lots of samples and step-by-step tutorials. The APS GitHub page contains many useful sample projects:
https://aps.autodesk.com/developer/documentation
https://aps.autodesk.com/developer/overview/manufacturing-data-model-api
https://aps.autodesk.com/developer/overview/aec-data-model-api
https://github.com/autodesk-platform-services

#### APS Data Model Tutorials

The two tutorials for the Manufacturing and AEC Data Model APIs help developers to quickly understand both APIs:
https://autodesk-platform-services.github.io/aps-mfgdm-tutorial/
https://autodesk-platform-services.github.io/aps-aecdm-tutorial/

# Manufacturing Data Model API

Manufacturing Data Model Project: ERP Integration with Fusion




### Project Overview

- ERP System: Microsoft Business Central
- Data Managed: Component structure, quantities, properties, and BOMs
- On-demand Transfer: Real-time transfer and updates between Fusion and Business Central using GraphQL and OData for ERP integration
- Customer Requirements: Ensuring accurate and up-to-date transfer of items and BOMs to Business Central, including automated part number generation
- Solution:
  - GraphQL Queries: Retrieve component hierarchy, material properties, and thumbnails
  - OData: Used for data comparison and updates in the ERP system

### Sample Code - Installation and Configuration

The project consist of two components: an Autodesk Fusion Add-In and a .NET Core web application. The project also makes use of a „Custom Property" with the name "ERP Number" to persist part numbers created in the ERP system on the Fusion component. Here's how to set it up:

### Part 1: Install the Fusion Plugin

- Download or clone the GitHub repository: https://github.com/christiangessner/AU2024_SD2336
- Open Autodesk Fusion CAD
- Open “Scripts and Add-Ins…” from Utilities – Add-Ins
- Click the green “+” button next to “My Add-Ins”
- Select the downloaded folder “*Use Case - Manufacturing Data Model (Fusion AddIn)*” and click “Open”
- Click “Run” to load the Add-In

### Part 2: Create an APS Application

- Create an APS Application of type “Traditional Web App” as explained here
  - https://aps.autodesk.com/en/docs/oauth/v2/tutorials/create-app/
- When creating the app, from the “API Access” dropdown, select
  - Manufacturing Data Model API
  - Parameters API
- For the Callback URL, provide a local URL, e.g., `http://localhost:8080/api/auth/callback`
- Remember “Client ID”, “Client Secret” and “Callback URL” from the new app

### Part 3: Configure Business Central

Go to the chapter “Microsoft Business Central - Configuration” and complete all steps before proceeding to the next step. Remember “Client ID”, “Client Secret” from these steps and find out your Business Central “Tenant ID”, “Environment” and “Company”.

### Part 4: Configure and run the .NET Core Web Application

- Download or clone the GitHub repository: https://github.com/christiangessner/AU2024_SD2336
- Open Visual Studio Code (VS Code)
- In VS Code, open the folder “Use Case - Manufacturing Data Model (Web App)”
- In the VS Code Explorer, find and open the file “*appsettings.json*”
- Provide details of the APS application (Part 2) for the settings:
  - APS_CLIENT_ID
  - APS_CLIENT_SECRET
  - APS_CALLBACK_URL
- Provide details of the Business Central application (Part 3) for the settings:
  - BC_CLIENT_ID
  - BC_CLIENT_SECRET
  - BC_TENANT_ID
  - BC_ENVIRONMENT
  - BC_COMPANY

The following tutorial shows how the create a custom "Property Definition" to be used for Fusion components: https://autodesk-platform-services.github.io/aps-mfgdm-tutorial/properties/home/.
Follow the tutorial to create a new "Property Definition Collection" and a "Property Definition":

- Name: "ERP Number"
- Type "String"
- Behavior: "Dynamic"

To start the application in VS Code, go to "Run and Debug" or press "F5"

## Microsoft Business Central - Configuration

### Part 1: Enable OAuth for Business Central in Azure

OAuth 2.0 is the protocol used to authenticate apps with Business Central. Here's how to set it up in Azure Active Directory (AAD):

**Step 1: Create an App Registration in Azure**

- Go to the Azure Portal (https://portal.azure.com)
- In the portal menu, select "Microsoft Entra ID"
- Under "Manage", choose "App registrations", then click "New registration"
- Fill in the form:
  - Name: Choose a meaningful name (e.g., "Business Central OAuth App").
  - Supported account types: Choose "Accounts in this organizational directory only"
- Click "Register"

**Step 2: Configure API Permissions**

- After registration, go to your newly created app
- In the left-hand panel, under "Manage", click "API permissions", then "Add a permission"
- Select "Microsoft APIs", and search for "Dynamics 365 Business Central"
- Choose the "Delegated permissions": "user_impersonation for common access"
- Click "Add permissions"
- You may need to click "Grant admin consent" for the permissions to be effective

**Step 3: Generate Client ID and Secret**

- In the left panel, select "Certificates & Secrets"
- Under "Client secrets", click "New client secret"
- Add a description (e.g., "Business Central Secret") and choose an expiration date
- Click "Add"
- Copy the "Client Secret" and "Client ID"

**Step 4: Configure Business Central**

- Log into Business Central
- Go to “Microsoft Entra Applications” under “Administration”
- Click “New”, then enter the “Client ID” from Azure
- Select “User Assignment” and choose the permissions needed
- In “User Permissions”, define which users will have access

## Part 2: Enable Web Services in Business Central

Business Central supports web services like SOAP, OData, and APIs. Here’s how to enable and configure OData web services:

**Step 1: Enable Web Services**

- Open Business Central and navigate to the search bar (top-right)
- Search for “Web Services” and select the result under the “Administration” section
- In the “Web Services” page, click “New” to create a new service

**Step 2: Set Up the Web Service**

- Choose the service type: OData
- Specify the object to expose: Page
- Define the Object ID (this refers to the ID of the page)
- Enter a Service Name (this will form part of the service URL)
- Set the “Enabled” checkbox to “Yes”

The following Web Services must be added for the sample code to work:

| Object Type | Object ID | Object Name | Service Name (*) |
|---|---|---|---|
| Page | 30 | Item Card | ItemCards |
| Page | 30008 | APIV2 - Items | Items |
| Page | 99000786 | Production BOM | ProductionBOMs |
| Page | 99000788 | Lines | ProductionBOMLines |

** Make sure, the web services are "Published" and the **Service Name** exactly matches the value from the table. If not, the web app cannot find the URL and terminates with errors!*

**Step 3: Access the Web Service**

- After setting up, the service URL will be visible in the Web Services list
- You can access the service by going to:
  `https://api.businesscentral.dynamics.com/v2.0/{TENANT_ID}/ODataV4/{ServiceName}`
- Use an HTTP client like Postman or your application to interact with the service

## Microsoft Business Central - Useful Links

**Microsoft Business Central Documentation**
Official documentation covering everything from setup to advanced integrations:
https://docs.microsoft.com/en-us/dynamics365/business-central/

**Business Central OAuth 2.0 Authentication**
Step-by-step guide for setting up OAuth authentication in Business Central:
https://docs.microsoft.com/en-us/dynamics365/business-central/dev-itpro/administration/setting-up-azure-active-directory

**Microsoft Azure Portal**
The portal where you manage Azure resources, including app registrations for OAuth:
https://portal.azure.com/

**API Documentation for Business Central**
Detailed documentation on how to use APIs to interact with Business Central:
https://docs.microsoft.com/en-us/dynamics365/business-central/dev-itpro/api-reference/v2.0/

**OData Web Services in Business Central**
Information about enabling and using web services in Business Central:
https://docs.microsoft.com/en-us/dynamics365/business-central/dev-itpro/webservices/web-services

## AEC Data Model API

AEC Data Model Project: Automating Deliveries to Odoo (Inventory module)

```
C:\Windows\System32\Windc

Round Duct / Tees 4"ø (120")                           84    74      518.1 11.7", 9.8", 9.3", 8.7", 8...
Round Duct / Tees 4"ø (240")                           67    67      705.2 11.2", 6.5", 6.5", 6.5", 6...
Round Duct / Tees 4"ø (480")                           10    10      194.5 24", 22.8", 20.5", 18.9", ...
Round Duct / Tees 6"ø (12")                            17    16       56.5
Round Duct / Tees 6"ø (24")                             9     7       33.5
Round Duct / Tees 6"ø (60")                             4     4         32 11", 11"
Round Duct / Tees 6"ø (120")                            3     3       27.7 9.2", 9.2", 9.2"
Round Duct / Tees 8"ø (24")                            76    14      276.8 6"
Round Duct / Tees 8"ø (48")                            49    12      153.3 6", 6"
Round Duct / Tees 8"ø (60")                            36     1      207.8 9.6", 9.6", 9.6", 9", 8.8"...
Round Duct / Tees 8"ø (120")                          100    87      488.4 7", 7", 7", 6", 6", 9.4", ...
Round Duct / Tees 8"ø (240")                           44    44      351.9 11.2", 11.2", 11.2", 9.8",...
Round Duct / Tees 8"ø (480")                           16    16      158.1 21.3", 21.3", 21.3", 17", ...
Round Elbow / 1 D 4"ø-4"ø                              28    28
Round Elbow / 1 D 6"ø-6"ø                              10    10
Round Elbow / 1 D 8"ø-8"ø                               9     9
Round Elbow / 1.5 D 4"ø-4"ø                           550   550
Round Elbow / 1.5 D 6"ø-6"ø                            16    16
Round Elbow / 1.5 D 8"ø-8"ø                           319   319
Round Endcap / Standard 6"ø                             3     3
Round Endcap / Standard 8"ø                           100   100
Round Tee / Standard 8"ø-8"ø-6"ø                       24    24
Round Tee / Standard 8"ø-8"ø-8"ø                       36    36
Round Transition - Angle / 45 Degree 4"ø-4"ø            2     2
Round Transition - Angle / 45 Degree 8"ø-6"ø            9     9


Total waste including re-usage of leftovers: 5027"
51 unique ducts found in 2 file(s)!
```

### Project Overview

- ERP System: Odoo (Inventory module)
- Data: Ducts and fittings from Revit designs, optimized for delivery and inventory management
- Automated Deliveries: Autodesk Construction Cloud submittals trigger automated creation of deliveries in Odoo
- Solution:
    - GraphQL Queries: Pull relevant data such as dimensions and quantities from Revit models
    - Odoo API (JSON-RPC): Automates product creation and delivery scheduling

### Sample Code - Installation and Configuration

The project uses the Windows Task Scheduler to run a PowerShell script. The script authenticates with APS, reads Submittals from Autodesk Construction Cloud (ACC), finds Revit files that are referenced with the Submittals and then uses the AEC Data Model API to read ducts and fittings from these files. With this data the script automatically creates products and deliveries in the ERP system Odoo (Inventory module). Here’s how to set it up:

### Part 1: Create an APS Application

- Create an APS Application of type “Traditional Web App” as explained here
  - [https://aps.autodesk.com/en/docs/oauth/v2/tutorials/create-app/](https://aps.autodesk.com/en/docs/oauth/v2/tutorials/create-app/)
- From the “API Access” dropdown list, select
  - Manufacturing Data Model API
  - Parameters API
- For the Callback URL, provide a local URL, e.g., `http://localhost:8080/api/auth/callback`
- Remember “Client ID”, “Client Secret” and “Callback URL” from the new app

### Part 2: Configure ACC API Access

- Provide access to your client ID in your ACC account by following the steps mentioned in the Provisioning access in other products page in the APS documentation

### Part 3: powerAPS (3LO APS authentication without login box)

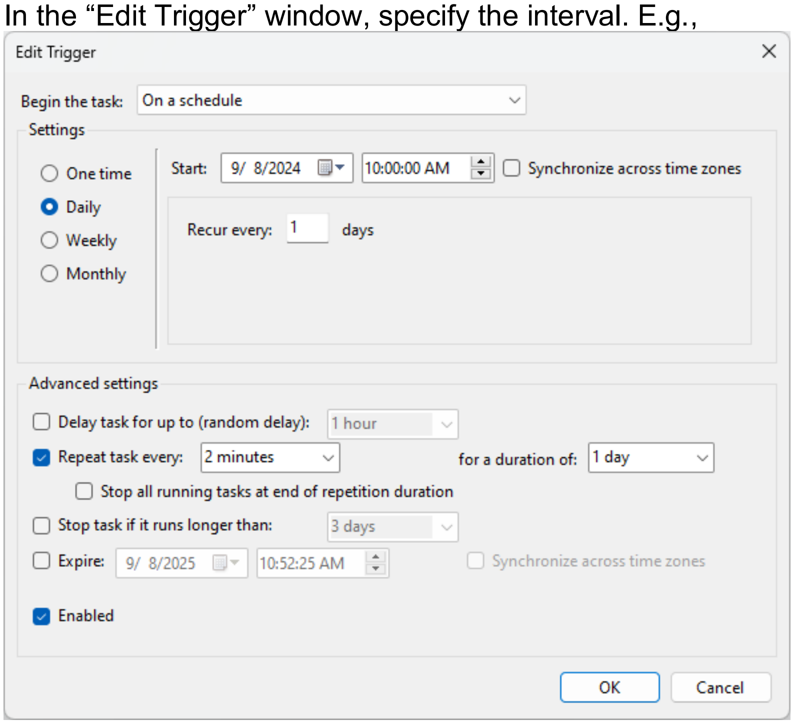
- Download the installer from the powerAPS GitHub repository: [https://github.com/coolOrangeLabs/powerAps/releases](https://github.com/coolOrangeLabs/powerAps/releases)
- Run the MSI-installer and follow the setup wizard
- Click “OK” to finish the setup

### Part 4: PowerShell Script

- Download or clone the GitHub repository: [https://github.com/christiangessner/AU2024_SD2336](https://github.com/christiangessner/AU2024_SD2336)
- Open Visual Studio Code (VS Code)
- In VS Code, open the folder “*Use Case - AEC Data Model*”
- In the VS Code Explorer, find and open the file “*ACC Ducts Export to Odoo.ps1*”
- In the region “Settings” (line 3 to line 31) set the following variables:
  - `$hubName` = “<Name of your ACC Hub>”
  - `$projectName` = “<Name of your ACC Project>”
  - `$clientId` = “<Client ID generated in Part 1>”
  - `$clientSecret` = “<Client Secret generated in Part 1>””
  - `$callbackUrl` = “<Callback URL generated in Part 1>””
  - `$username` = “<Autodesk ID (email address)>”
  - `$password` = “<Autodesk ID password>”
  - `$odooDatabase` = “<Name of your Odoo Database>”
  - `$odooUsername` = “<Odoo username (email address)>”
  - `$odooPassword` = “<Odoo password>”
- Save the changes and remember the location of the file for the next part (Windows Task Scheduler)

## Part 3: Windows Task Scheduler

- From the Windows start menu, open the “Task Scheduler”
- From the Action menu, click “Create Basic Task…”
- Specify a Name for the Task, e.g.: “Ducts from ACC to Odoo”, click “Next >”
- Set “Daily” on the task trigger tab, then click “Next >”
- Specify a start day and time and an interval, e.g.: 1 day, then click “Next >”
- Select “Start a program”, click “Next >”
- Specify the “Program/script” and “Add arguments”, then click “Next >”
    - Program/script:
      ```
      C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
      ```
    - Add arguments:
      ```
      -ExecutionPolicy Bypass -NonInteractive -WindowStyle Hidden
      -File "<Your Script File location>\ACC Ducts Export to
      Odoo.ps1"
      ```
- Select “Open the Properties dialog for this task when I click Finish” and click “Finish”
- On the “General” tab, select “Run only when user is logged on” and check “Run with highest privileges”
- On the “Triggers” tab, select “Edit…”
- In the “Edit Trigger” window, specify the interval. E.g.,




- Click “OK” to change the trigger to your individual needs
- Click “OK” to save and close the task

The Windows Task Scheduler runs the PowerShell script in the configured interval. In the context menu of the newly created task, the command “Run” can be used to test the script.

### Odoo - Account

Creating Free Account and Using JSON-RPC

### Create a Free Odoo Account

- Go to the Odoo website: https://www.odoo.com/.
- Click on Try it free in the top-right corner
- Choose the Odoo apps you want to try (you can select more apps later)
- Fill out your Company Name and Email, then click Start Now
- Odoo will generate a cloud-hosted instance for you, with a unique URL (e.g., yourcompany.odoo.com)
- Check your email to confirm your registration
- Log in to your Odoo instance using your email and password

### Odoo - Useful Links

**Odoo Official Website**
This is where you can sign up for a free trial and explore Odoo's features:
https://www.odoo.com/

**Odoo Developer Documentation**
Comprehensive documentation for developers, including JSON-RPC and API integrations:
https://www.odoo.com/documentation/16.0/developer.html

**Odoo Web Services (JSON-RPC) Documentation**
Detailed explanation of how to use JSON-RPC to interact with Odoo's web services:
https://www.odoo.com/documentation/16.0/webservices/odoo.html

**Odoo GitHub Repository**
Access the source code and community modules on GitHub for custom development:
https://github.com/odoo/odoo

**Odoo API Authentication**
Learn how to set up authentication for Odoo's APIs using JSON-RPC:
https://www.odoo.com/documentation/16.0/developer/howtos/authentication.html